

Euler's Gem

(V, E) finite simple graph

Inductive definitions:

G is contractible: exists v , such that $S(v)$ and $G-v$ are contractible. 1 is contractible.

G is q-manifold: all $S(v)$ are (q-1)-spheres. G is q-sphere: exists v such that $G-v$ is contractible. 0 is the (-1)-sphere.

Euler's Gem

If G is a q-sphere, then

$$\chi(G) = 1 + (-1)^q$$



Descartes



Euler



Schläfli

Lemmas

Every $B(v)$ is contractible.

Proof: Induction. Take $w \in V(B) \setminus v$. Then $B(w)$ is smaller and so contractible. $S(w)$ is unit ball in a smaller graph.

G contractible implies $\chi(G) = 1$

Proof: induction with $n=|V|$. Pick v such that $G \setminus v$ is contractible $\chi(G) = \chi(G \setminus v) + \chi(B(v)) - \chi(S(v)) = 1 + 1 - 1$

$$\chi(A \cup B) + \chi(A \cap B) = \chi(A) + \chi(B)$$

Proof: Use linearity and check it for valuations $A \rightarrow f_k(A)$

Proof of Gem

If G is a q-sphere, then

$$\chi(G) = 1 + (-1)^q$$

Proof: Induction. Check $q=0$.

Pick v such that $G-v$ is contractible.

Use

$$\chi(G) = \chi(G \setminus v) + \chi(B(v)) - \chi(S(v))$$

Fill in the inductive assumption.

$$\chi(S(v)) = 1 - (-1)^q$$

Use

$$\chi(G \setminus v) = 1$$

and

$$\chi(B(v)) = 1$$

This gives $\chi(G) = 1 + 1 - (1 + (-1)^{q-1})$

Small q

If G is a q-sphere, then

$$\chi(G) = 1 + (-1)^q$$

q=-1

$$\chi(G)=0$$

q=0

$$|V|=2$$

$$\chi(G)=2$$

q=1

$$|V|=10$$

$$|E|=10$$

$$\chi(G)=10-10=0$$

q=2

$$|V|=32$$

$$|F|=60$$

$$|E|=90$$

$$\chi(G)=32-90+60=2$$

q=3

$$|V|=8$$

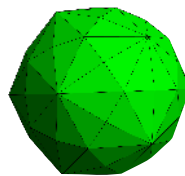
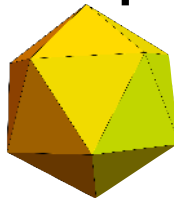
$$|F|=32$$

$$|E|=24$$

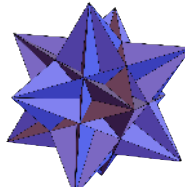
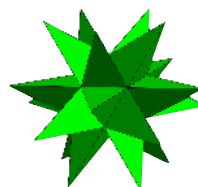
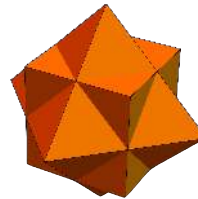
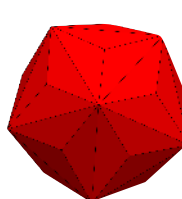
$$|C|=16$$

$$\chi(G)=8-24+32-16=0$$

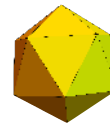
Spheres



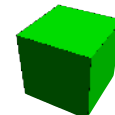
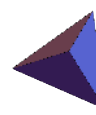
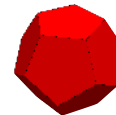
No Spheres



Plato



$$V - E + F = 2$$



$$\chi = 4 - 6 + 4 - 1 = 1$$

$$\text{maxdim}=3$$

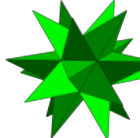
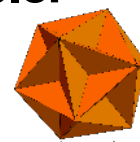
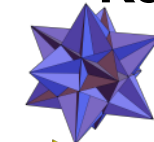
$$\chi = 20 - 30 = -10$$

$$\text{maxdim}=1$$

$$\chi = 8 - 12 = -4$$

$$\text{maxdim}=1$$

Kepler



Epistemology



Imre Lakatos 1922-1974

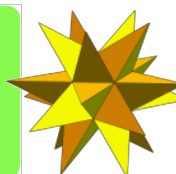
"Monster Barring" —

"Monster Adjustment"

"Monster Handling"

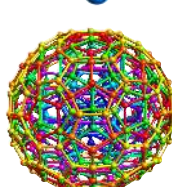
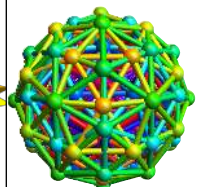
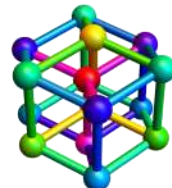
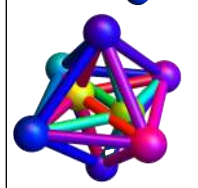
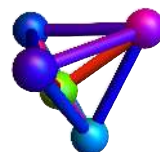
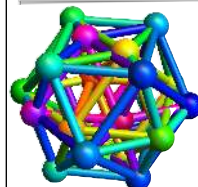


Better: avoid monsters by making precise, unambiguous and simple definitions!



Platonic Spheres

G is platonic q-sphere if it is a q-sphere and if there is a Platonic (q-1) sphere A, such that every $S(v)$ is equal to A.



Classification

∞ many Platonic 1-spheres

Proof: every connected 1-manifold is a circular graph $C(n)$, $n \geq 3$.

2 Platonic 2-spheres.

Proof: Gauss-Bonnet. Curvature must be positive and add up to 2.



2 Platonic 3-spheres.

Proof: Unit sphere must be one of the two $q=2$ Platonic spheres. Both work:

$$f=(120, 720, 1200, 600)$$



$$f=(8, 24, 32, 16)$$

1 Platonic 4-sphere.

Proof: Gauss-Bonnet. Curvature of a 4-manifold is $K = 1 - E/6 + F/10$

$$K=1$$

$$K=1/5$$

Being down to 1, there is only one Platonic sphere for all $q \geq 3$.